

Cognitive Science — Module 01: Systems — Study Questions

1. What is a system? Give three examples of systems from vastly different domains (biology, technology, society). What do they have in common?
2. What is the difference between an open system and a closed system? Why are living things necessarily open systems? What would happen if they became closed?
3. In your own words, explain what a Markov Blanket is. Why is it called a "blanket"? Draw one for a cell.
4. What are the four types of states in a Markov Blanket partition? Give an example of each for a human being, a cell, and a university.
5. What does it mean that internal states and external states are "conditionally independent given the blanket states"? Explain this in plain English using the skin analogy.
6. What is emergence? Why can't you understand a system by studying its parts in isolation? Give an example where the whole is dramatically different from the sum of its parts.
7. Compare homeostasis and allostasis. Give two real-world examples of each. Why is the distinction important for understanding cognition?
8. What does it mean for a living system to be "far from equilibrium"? What is the alternative (equilibrium), and why is it equivalent to death?
9. How does a thermostat perform homeostasis? In what ways is a thermostat a limited version of an agent? What does it lack?
10. How does the concept of a Markov Blanket apply at multiple scales? Can a nation have a Markov Blanket? What about a social group?
11. Give an example of allostasis from your own life today. How did you anticipate a future need and prepare for it in advance?
12. Why is the second law of thermodynamics relevant to living systems? How do organisms resist the tendency toward disorder?
13. What is the connection between the Markov Blanket concept and the idea of personal identity? Is "you" defined by the matter inside your blanket, or by the blanket itself?
14. How might climate change be understood as a disruption of a planetary system's homeostasis? What are the "prediction errors" in this analogy?

15. Critically evaluate: Is the Markov Blanket always a clear, sharp boundary, or can it be fuzzy, nested, or overlapping? Give examples that challenge the "clear boundary" assumption.
16. How does the concept of "autopoiesis" (self-making) relate to the Markov Blanket? An autopoietic system continuously produces and replaces its own components. Explain how a cell membrane operates as both a physical boundary and a statistical Markov Blanket simultaneously.
17. What is the difference between robustness and resilience in a system? A robust system resists perturbation; a resilient system recovers from it. Give a biological example of each and explain how Active Inference might account for both.
18. How does thermodynamics relate to systems theory? What does the Second Law of Thermodynamics predict about any system, and how does the Free Energy Principle explain why living systems appear to resist this tendency toward disorder?
19. Consider a city as a system. What are its internal states, sensory states, and active states? Where would you draw its Markov Blanket? Does viewing a city through Active Inference generate any non-obvious insights about urban planning?
20. How does a system's complexity relate to the richness of its Markov Blanket? Compare the Markov Blanket of a rock, a bacterium, a mouse, and a human. What increases as we move up this hierarchy, and what fundamental property of each system does the Blanket complexity reflect?